

STUDYING GRAMMATICAL CHANGE USING THE PENN PARSED CORPORA OF HISTORICAL ENGLISH

George Walkden • <http://walkden.space>

University of Konstanz

ISLE Summer School, Uppsala, June 2026

AIMS OF THE COURSE

- This workshop will provide a hands-on introduction to using **richly annotated historical corpora** for the study of language change.
- After a brief overview of
 - the available corpora and
 - the sort of questions that can be answered
- we'll use **parsed texts** from modern English (1500–) to formulate and run some simple queries using **CorpusSearch 2**.
- Participants will need to bring their **own laptops** with **Java** installed.

COURSE SCHEDULE

- Wed 10th 9–10:30 **Introduction, theory, and available corpora**
- Wed 10th 14:30–16:30 **Practical: Starting out with CorpusSearch**
- Thur 11th 9–10:30 **Practical: More advanced CorpusSearch queries**
- Thur 11th 11:00–12:00 **Case studies and workshop session**

CORPUS LINGUISTICS

- **Corpus** = a collection of naturally occurring linguistic data
- **Corpus linguistics** is a broad methodology for studying language, *not* a theory or branch of linguistics
- The approach I take is **corpus-based** rather than corpus-driven (Gries 2010): corpora can be useful for testing and improving theories
- For more on corpus linguistics in general, see McEnery & Wilson (2001) and Stefanowitsch (2020)

CHOMSKY'S CRITICISMS

- Focus of linguistics is (or should be) **competence**
- Corpora represent **performance**
- The syntax of natural languages is usually thought to offer **infinite** possibilities; corpora are by necessity **finite**
- “Any natural corpus will be skewed. Some sentences won’t occur because they are obvious, others because they are false, still others because they are impolite. The corpus, if natural, will be so wildly skewed that the description [of language based on the corpus] would be no more than a mere list.”

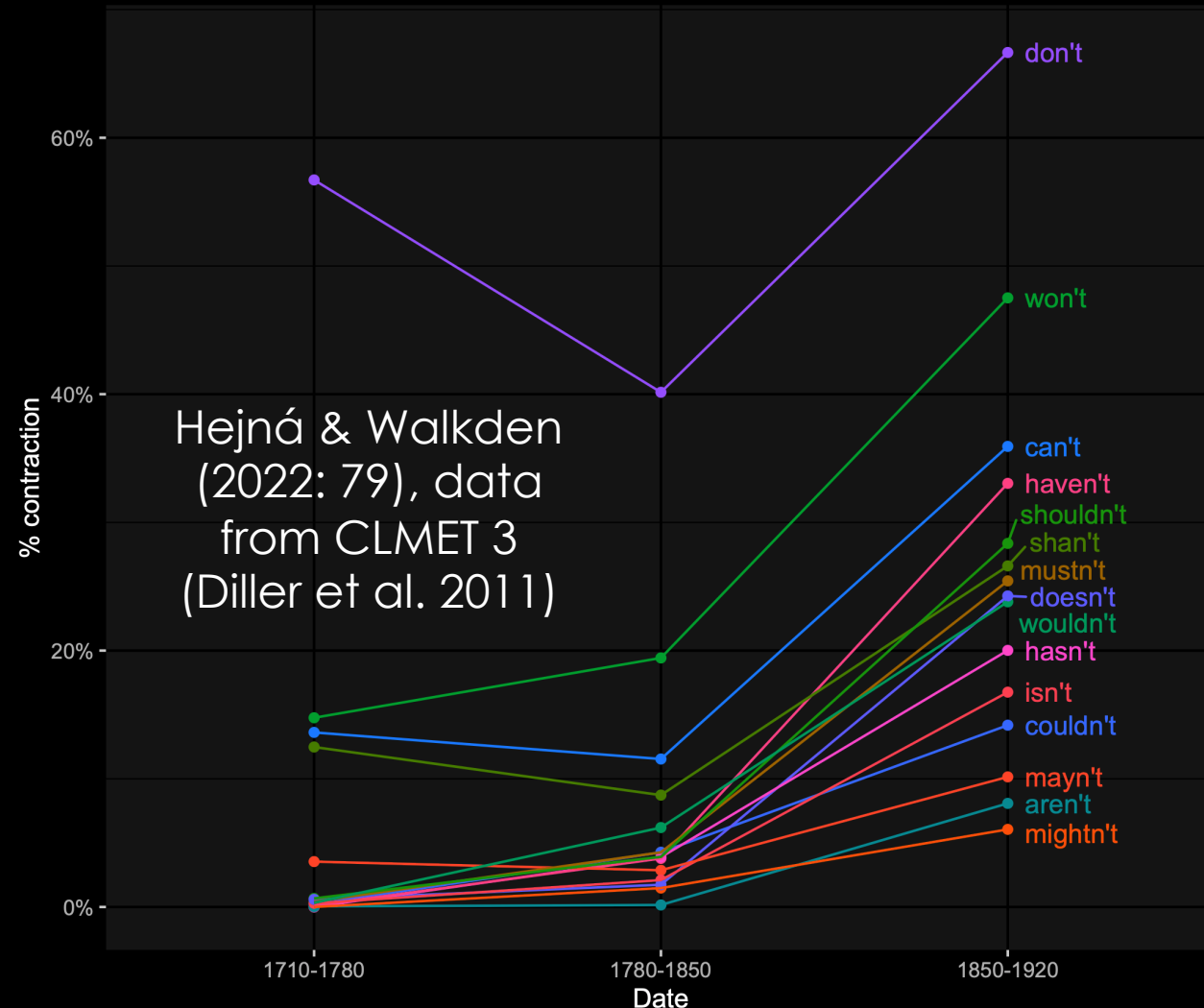
Chomsky (1962: 159), in McEnery & Wilson (2001: 10)

- Alternative methodology: **introspection**

USEFULNESS OF CORPORA

- To answer questions that fall outside of what Chomsky is/was interested in
 - NLP applications
 - Sociolinguistic questions
 - Questions about language change
 - Questions about performance *per se*
- In situations where other kinds of data are problematic or unavailable
 - Early child language production
 - Stigmatized varieties
 - Cases where intuitions are unclear or variable
 - Historically-attested languages and language change!
- Introspection and corpus linguistics are **complementary** (Fillmore 1992)

STUDYING LANGUAGE CHANGE



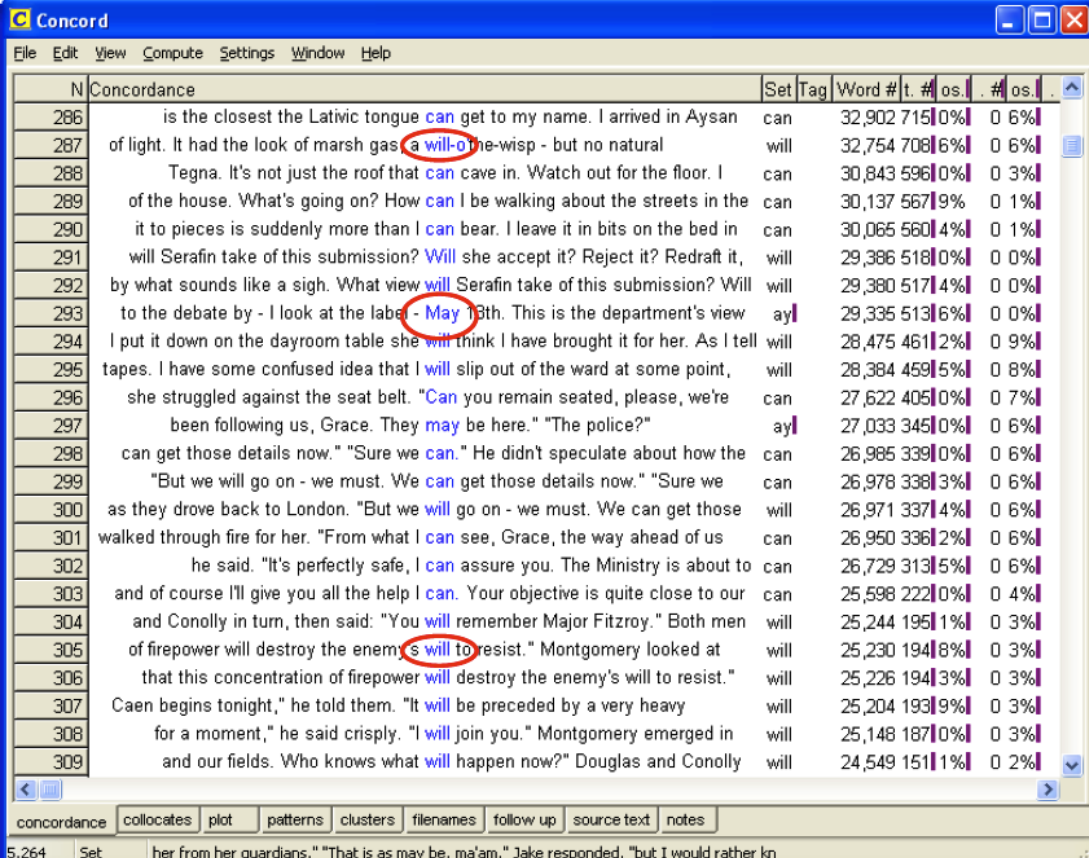
- At the community level, and at the text level, change involves variation and fluctuation of frequency
- We rarely have enough data to study change in an individual author (though see the EMMA corpus, Petré et al. 2019)
- Figure: Negative contraction as a proportion of use (e.g. *should not* vs. *shouldn't*) during the Late Modern English period

LIMITATIONS OF RAW CORPORA

- Raw corpora are limiting.
- Let's say you want to find the English modals *may*, *can* and *will*. What problems are likely to arise when searching for these?

LIMITATIONS OF RAW CORPORA

- Raw corpora are limiting.
- Let's say you want to find the English modals *may*, *can* and *will*. What problems are likely to arise when searching for these?



The screenshot shows the Concord software interface with a concordance search for the word 'can'. The search results are displayed in a table with columns for line number, concordance text, set, tag, word number, frequency, and percentage. The word 'can' is highlighted in blue in the concordance text. The search results are as follows:

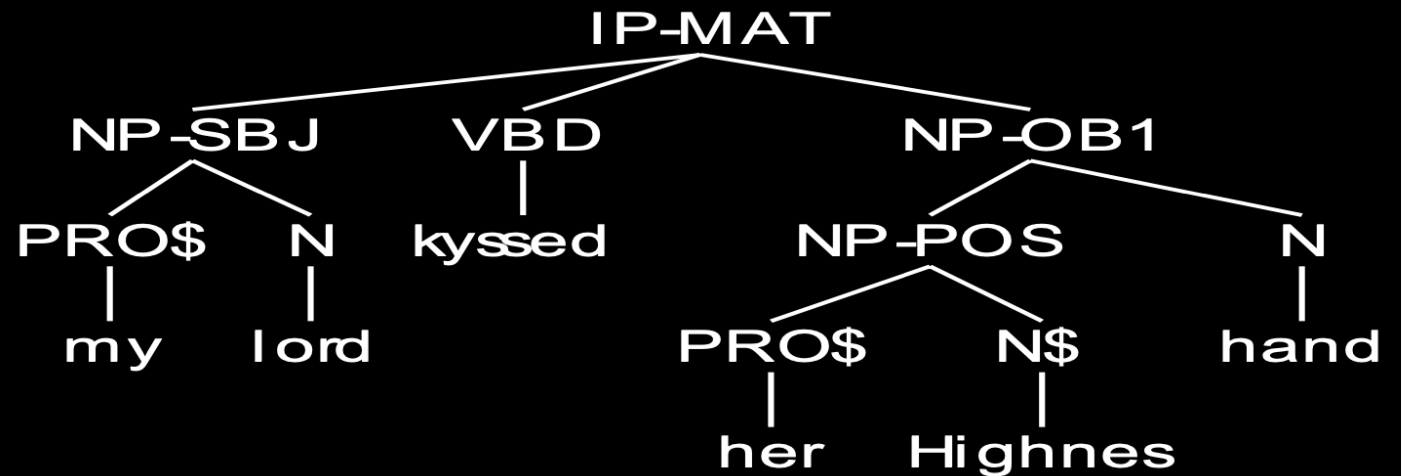
N	Concordance	Set	Tag	Word #	t. #	os. #	%
286	is the closest the Latic tongue can get to my name. I arrived in Aysan	can		32,902	715	0%	0 6%
287	of light. It had the look of marsh gas a will-o-the-wisp - but no natural	will		32,754	708	6%	0 6%
288	Tegna. It's not just the roof that can cave in. Watch out for the floor. I	can		30,843	596	0%	0 3%
289	of the house. What's going on? How can I be walking about the streets in the	can		30,137	567	9%	0 1%
290	it to pieces is suddenly more than I can bear. I leave it in bits on the bed in	can		30,065	560	4%	0 1%
291	will Serafin take of this submission? Will she accept it? Reject it? Redraft it,	will		29,386	518	0%	0 0%
292	by what sounds like a sigh. What view will Serafin take of this submission? Will	will		29,380	517	4%	0 0%
293	to the debate by - I look at the label - May 18th. This is the department's view	ay		29,335	513	6%	0 0%
294	I put it down on the dayroom table she will think I have brought it for her. As I tell	will		28,475	461	2%	0 9%
295	tapes. I have some confused idea that I will slip out of the ward at some point,	will		28,384	459	5%	0 8%
296	she struggled against the seat belt. " Can you remain seated, please, we're	can		27,622	405	0%	0 7%
297	been following us, Grace. They may be here." "The police?"	ay		27,033	345	0%	0 6%
298	can get those details now." "Sure we can ." He didn't speculate about how the	can		26,985	339	0%	0 6%
299	"But we will go on - we must. We can get those details now." "Sure we	can		26,978	338	3%	0 6%
300	as they drove back to London. "But we will go on - we must. We can get those	will		26,971	337	4%	0 6%
301	walked through fire for her. "From what I can see, Grace, the way ahead of us	can		26,950	336	2%	0 6%
302	he said. "It's perfectly safe, I can assure you. The Ministry is about to	can		26,729	313	5%	0 6%
303	and of course I'll give you all the help I can . Your objective is quite close to our	can		25,598	222	0%	0 4%
304	and Conolly in turn, then said: "You will remember Major Fitzroy." Both men	will		25,244	195	1%	0 3%
305	of firepower will destroy the enemy's will to resist." Montgomery looked at	will		25,230	194	8%	0 3%
306	that this concentration of firepower will destroy the enemy's will to resist."	will		25,226	194	3%	0 3%
307	Caen begins tonight," he told them. "It will be preceded by a very heavy	will		25,204	193	9%	0 3%
308	for a moment," he said crisply. "I will join you." Montgomery emerged in	will		25,148	187	0%	0 3%
309	and our fields. Who knows what will happen now?" Douglas and Conolly	will		24,549	151	1%	0 2%

CORPUS ANNOTATION

- Solution: **annotation!**
- Annotation is interpretative information added to an electronic corpus of spoken or written language.
- Annotation may be:
 - **Textual** (telling us about the texts themselves)
 - **Linguistic** (providing information about the language in the texts)
- A common type of annotation is **part-of-speech (POS) tagging**
- In this course we'll focus on a richer type of annotation: **syntactic parsing**

SYNTACTIC PARSING

- **Parsing** = representing syntactic structure (i.e. beyond the individual word in isolation)
- **Treebank** = a parsed corpus
- Two main types of parsed corpus:
 - **Constituency**
 - **Dependency** (e.g. the PROIEL corpora)
- Today's workshop focuses on constituency parsing.



THE PENN CORPORA

- Constituency-parsed corpora
- Original: the Penn Treebank (Taylor, Marcus & Santorini 2003)
 - 2+ million words of parsed present-day English, built 1989–96
- Anthony Kroch, Susan Pintzuk, Beatrice Santorini and Ann Taylor (and collaborators) then produced parsed versions of the Helsinki Corpus of English Texts (Kytö 1996), built 1984–91
- From 2000 onwards, parsed corpora of other languages in the same style have been produced
- See Taylor (2020) on treebanks in historical syntax

```
( (S (`` ``)
(S-TPC-1
(NP-SBJ (PRP He) )
(VP (MD must)
(VP (VB have)
(VP (VBN forgiven)
(NP (PRP me) ))))
(' ' ')
(, ,)
(NP-SBJ (NNP Henrietta) )
(VP (VBD murmured)
(S (-NONE-*T*-1) )
(PP (TO to)
(NP (DT the) (NN room) )))
(. .) ))
```

AVAILABLE CORPORA: ENGLISH (1/3)

- Penn Parsed Corpora of Historical English (PPCHE): <https://www.ling.upenn.edu/hist-corpora/>
- Penn-Helsinki Parsed Corpus of Middle English (PPCME2) (Kroch & Taylor 2000): <http://www.ling.upenn.edu/ppche-release-2016/PPCME2-RELEASE-4>.
- Penn-Helsinki Parsed Corpus of Early Modern English (PPCEME) (Kroch, Santorini & Delfs 2004): <http://www.ling.upenn.edu/ppche-release-2016/PPCEME-RELEASE-3>.
- Penn Parsed Corpus of Modern British English (PPCMBE2) (Kroch, Santorini & Dierani 2016): <http://www.ling.upenn.edu/ppche-release-2016/PPCMBE2-RELEASE-1>.

AVAILABLE CORPORA: ENGLISH (2/3)

- York-Toronto-Helsinki Parsed Corpus of Old English Prose (YCOE) (Taylor, Warner, Pintzuk & Beths 2003): <http://www-users.york.ac.uk/~lang22/YcoeHome1.htm>.
- York-Helsinki Parsed Corpus of Old English Poetry (YCOEP) (Pintzuk & Plug 2001): <http://www-users.york.ac.uk/~lang18/pcorpus.html>.
- York-Helsinki Parsed Corpus of Early English Correspondence (PCEEC) (Taylor, Nurmi, Warner, Pintzuk & Nevalainen 2006): <http://www-users.york.ac.uk/~lang22/PCEEC-manual/index.htm>.

AVAILABLE CORPORA: ENGLISH (3/3)

- Parsed Corpus of Middle English Poetry (PCMEP) (Zimmermann 2014–a): <https://pcmep.net/>.
- Parsed Linguistic Atlas of Early Middle English (PLAEME) (Truswell, Alcorn, Donaldson & Wallenberg 2019): <http://www.amc.lel.ed.ac.uk/amc-projects-hub/project/p-laeme-a-parsed-linguistic-atlas-of-early-middle-english/>.
- Audio-Aligned and Parsed Corpus of Appalachian English (AAPCAppE) (Tortora, Blanchette, Santorini & Diertani 2017): <https://www.aapcappe.org/>.

AVAILABLE CORPORA: GERMAN

- Corpus of Historical Low German (CHLG) (Booth, Breitbarth, Ecay & Farasyn 2020): <https://www.chlg.ugent.be/>.
 - Includes Heliand Parsed Database (HeliPaD) (Walkden 2016): <https://doi.org/10.5281/zenodo.4395040>.
- Septembertestament (Light 2015)
- Geneva Corpus of Early German (GeCEG) (Zimmermann 2014–b)

AVAILABLE CORPORA: OTHER GERMANIC

- Icelandic Parsed Historical Corpus (IcePaHC) (Wallenberg, Ingason, Sigurðsson & Rögnvaldsson 2011):
[https://linguist.is/icelandic_treebank/Icelandic_Parsed_Historical_Corpus_\(IcePaHC\)](https://linguist.is/icelandic_treebank/Icelandic_Parsed_Historical_Corpus_(IcePaHC)).
- Penn Parsed Corpus of Historical Yiddish (Santorini 2021):
<https://github.com/beatrice57/penn-parsed-corpus-of-historical-Yiddish>.
- Parsed Corpus of Scottish Correspondence (PCSC) (Gotthard 2022-):
<https://ulab.org.uk/conferences/conferences/45/presentations/550>.

AVAILABLE CORPORA: CELTIC

- Parsed Old and Middle Irish Corpus (POMIC) (Lash 2014): <https://www.dias.ie/celt/celt-publications-2/celt-the-parsed-old-and-middle-irish-corpus-pomic/>.
- Parsed Historical Corpus of the Welsh Language (PARSHCWL) (in preparation, see Meelen & Willis 2022): <https://www.celticstudies.net/parshcwl/>.

AVAILABLE CORPORA: FRENCH

- Modéliser le changement: Les voies de français (MCVF) (Martineau et al. 2021)
- Penn-BFM Parsed Corpus of Historical French (Kroch & Santorini 2021)
- Both available via <https://github.com/beatrice57/mcvf-plus-ppchf>.

AVAILABLE CORPORA: IBERO-ROMANCE

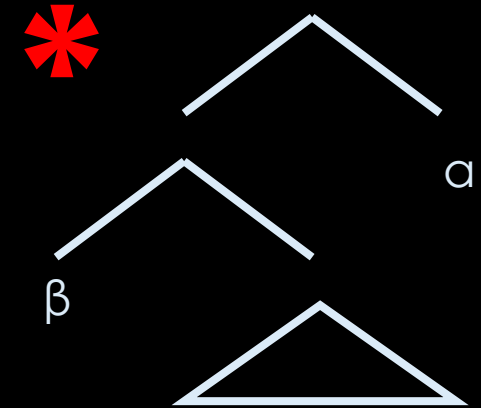
- Tycho Brahe Corpus of Historical Portuguese (Galves et al. 2017): <https://www.tycho.iel.unicamp.br/corpus/>.
- Word Order and Word Order Change in Western European Languages (WOChWEL, Portuguese): <http://alfclul.clul.ul.pt/wochwel/index.html>.
- Post Scriptum (historical Spanish and Portuguese letters): <https://hdl.handle.net/21.11129/0000-000D-F924-2>.
- Syntax-oriented Corpus of Portuguese Dialects (CORDIAL-SIN) (Martins 2000–): <http://www.clul.ulisboa.pt/en/10-research/314-cordial-s>.

OTHER AVAILABLE CORPORA

- Penn Parsed Corpora of Historical Greek (PPCHiG) (Beck 2013): <https://www.ling.upenn.edu/~janabeck/greek-corpora.html>.
- Oxford NINJAL Corpus of Old Japanese (ONCOJ): <https://oncoj.ninjal.ac.jp/>.
- Chinese Parsed Historical Corpus (ChiParHC) (Li 2017).
- Old Tibetan Treebank (in preparation, see Faggionato & Meelen 2019).

CASE STUDY 1: THE FINAL-OVER-FINAL CONDITION

- Biberauer, Holmberg & Roberts (2014), Sheehan et al. (2017):
 - **Final-over-Final Condition**
A head-final phrase aP cannot immediately dominate a head-initial phrase βP , if a and β are members of the same extended projection.
- The Final-over-Final Condition rules out Verb-Object-Auxiliary (VOAux) orders
 - e.g. *he eaten_V cake_O has_{Aux}*
- This prediction is right for Finnish, Dutch, German, Afrikaans, Basque... Is it right for Old English?

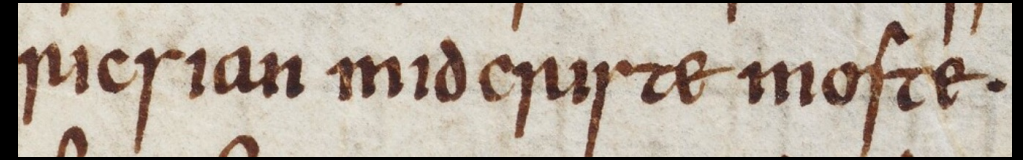


*VERB-OBJECT-AUXILIARY IN OLD ENGLISH

- In Old English, when the object is adjacent to the verb, 3 of 4 logically possible orderings of V[erb], O[bject] and Aux[iliary] are attested:
(Pintzuk 2005; evidence from YCOE, Taylor et al. 2003)
 - a) Se wolde gelytlian þone lyfigendan hælend (AuxVO)
he would diminish the living saviour
'he would diminish the living saviour' (colwstan1,+ALet 2 [Wulfstan 1]:188.256)
 - b) gif we ænige bote gebidan sculon (OVAux)
if we any relief expect must
'if we must expect any relief' (cowulf,WHom 20.2:18.1649)
 - c) Ac he sceal þa sacfullan gesibbian (AuxOV)
but he must the quarrelsome reconcile
'but he must reconcile the quarrelsome' (colwstan1,+ALet 2 [Wulfstan 1]:188.256)
 - d) *Ac he gesibbian þa sacfullan sceal (VOAux)

*VERB-OBJECT-AUXILIARY IN OLD ENGLISH

- Walkden (2014: 315): searched the YCOE for material intervening between a non-finite verb and a finite auxiliary in the same clause.
- Four clear examples, e.g.
 - *hu hie gedon ymbe þa menn hæfdan*
how they done about the men had
'... how they had dealt with the men'
(cobede, Bede 5:11.416.25.4189)
- None of these examples contained a nominal object between V and Aux.

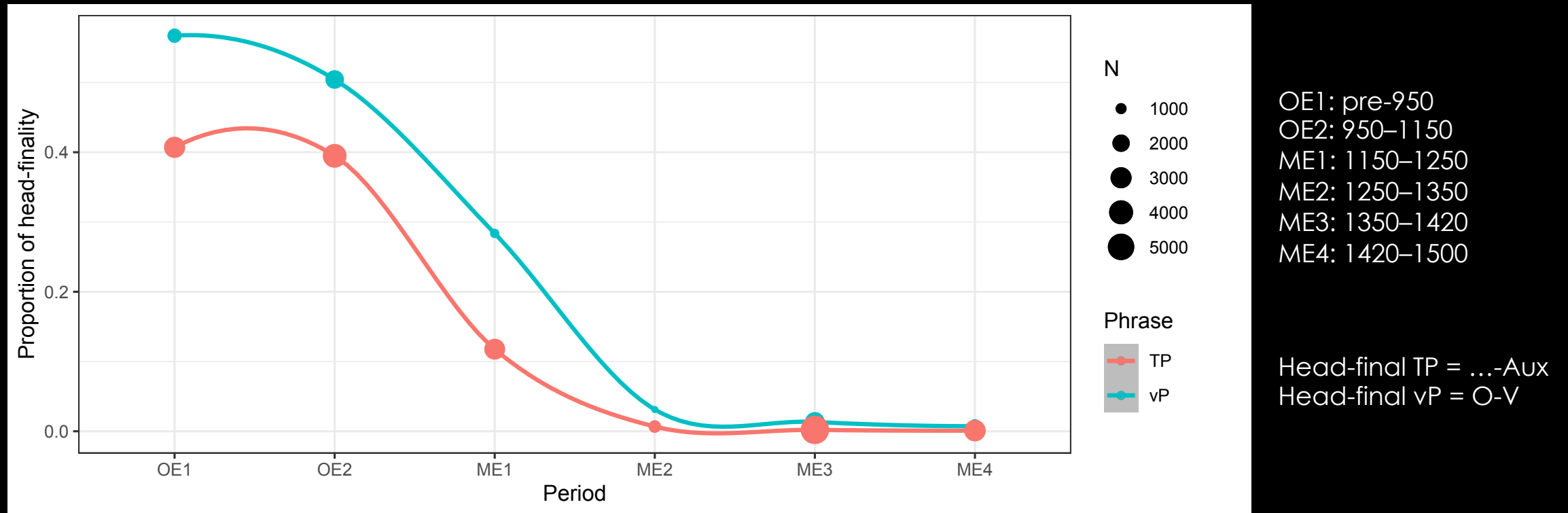


MS T, f. 057r (https://medieval.bodleian.ox.ac.uk/catalog/manuscript_8966)

VERB-PP-AUXILIARY IN BEDE

- V-XP-Aux is not quite unattested in Old English (Walkden 2014: 315–6, fn. 2)
 - a) þæt he swa ... ricsian mid Criste moste
that he so reign with Christ might
'that he may thus ... reign with Christ' (cobede,Bede:3:21.248.21.2544)
- There are 10 such examples (4 clear, 6 less clear cases)
 - Two are from the translation of Bede's *Historia*
 - But the five extant manuscripts diverge in constituent order! (Miller 1898)
 - Only MS T has V-XP-Aux for a); MSS B, Ca, C and O all show XP-V-Aux
 - Miller (1890: xiv) describes the scribe for this section of T as 'the most illiterate' of the five scribes working on the MS
 - Take-home message: apparent corpus counterexamples should be interpreted with philological-palaeographical tact

THE SHIFT TO HEAD-INITIAL (SVO)



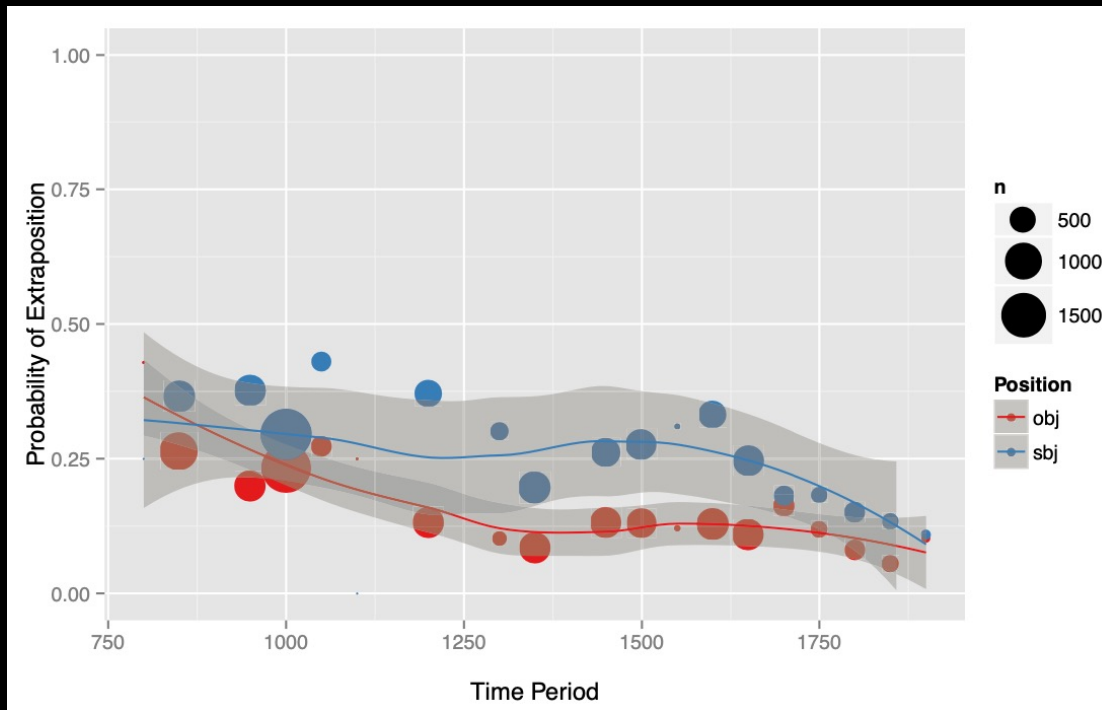
- Data sources: Pintzuk & Taylor (2006) for the vP, own search for TP using YCOE (Taylor et al. 2003) and PPCME2 (Kroch, Taylor & Santorini 2000)

CASE STUDY 2: RELATIVE CLAUSES

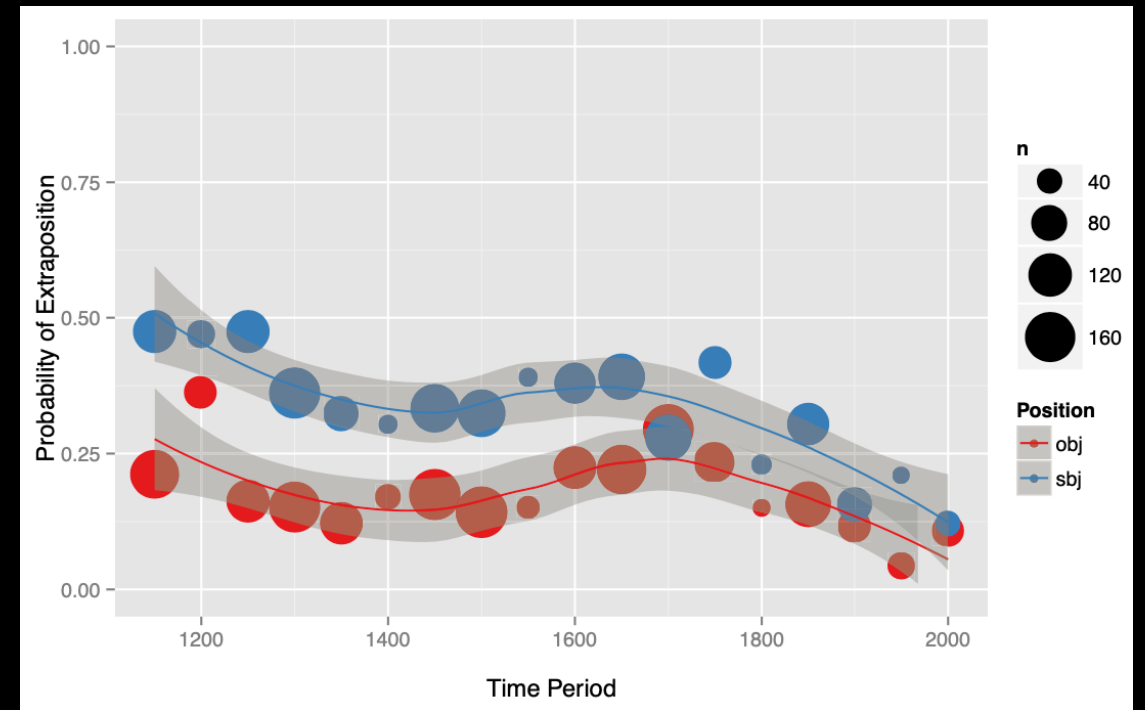
- Wallenberg (2016) investigates relative clause positioning over time across four languages: English, French, Icelandic and Portuguese.
- Variable of interest: whether or not the relative clause is **extraposed**.
 - **Another method** ... may be employed **which is less open to the above objection** (extraposition)
 - **Another method which is less open to the above objection** may be employed (no extraposition)
- Assumption of generative variationist syntax: these two variants reflect “competing grammars” (Wallenberg et al. 2026).

RELATIVE CLAUSE EXTRAPOSITION: RESULTS (ENGLISH, ICELANDIC)

- Extraposition gradually gets less and less frequent in all four languages.



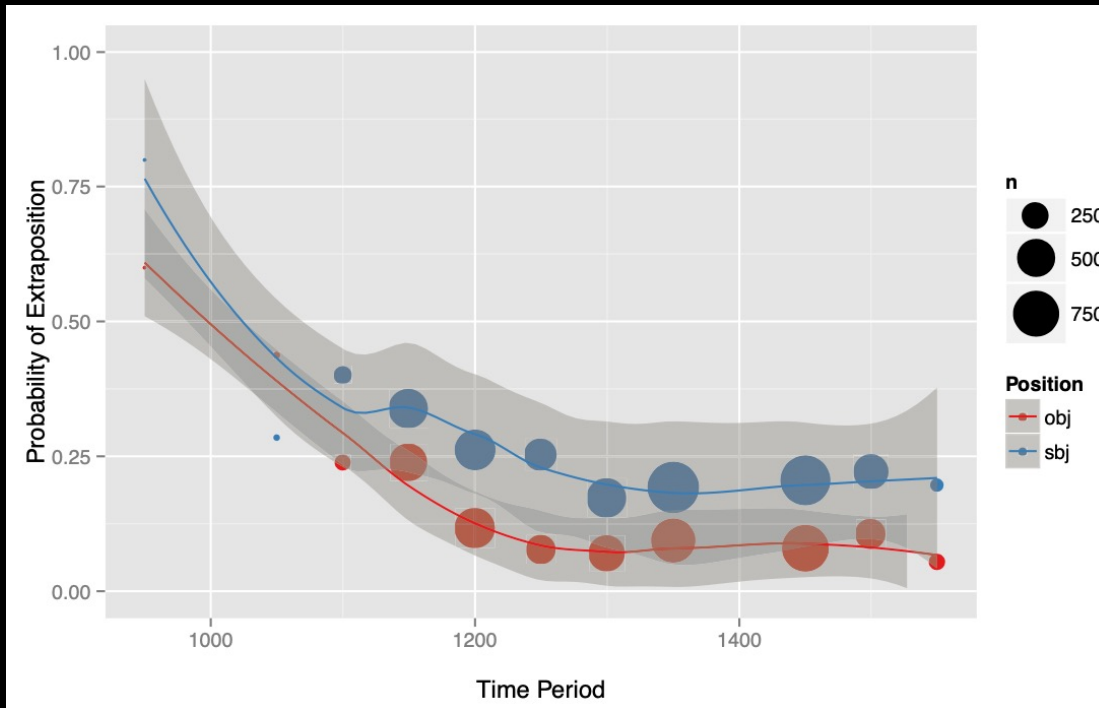
English



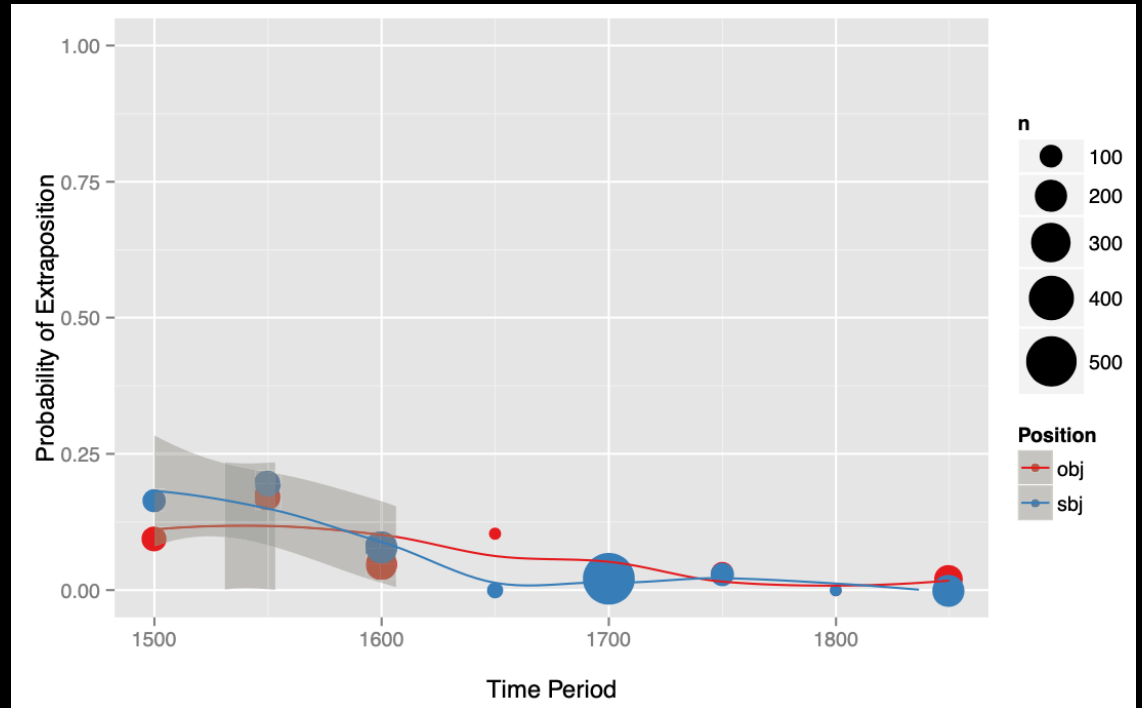
Icelandic

RELATIVE CLAUSE EXTRAPOSITION: RESULTS (FRENCH, PORTUGUESE)

- Extraposition gradually gets less and less frequent in all four languages.



French



Portuguese

RELATIVE CLAUSES: IMPLICATIONS

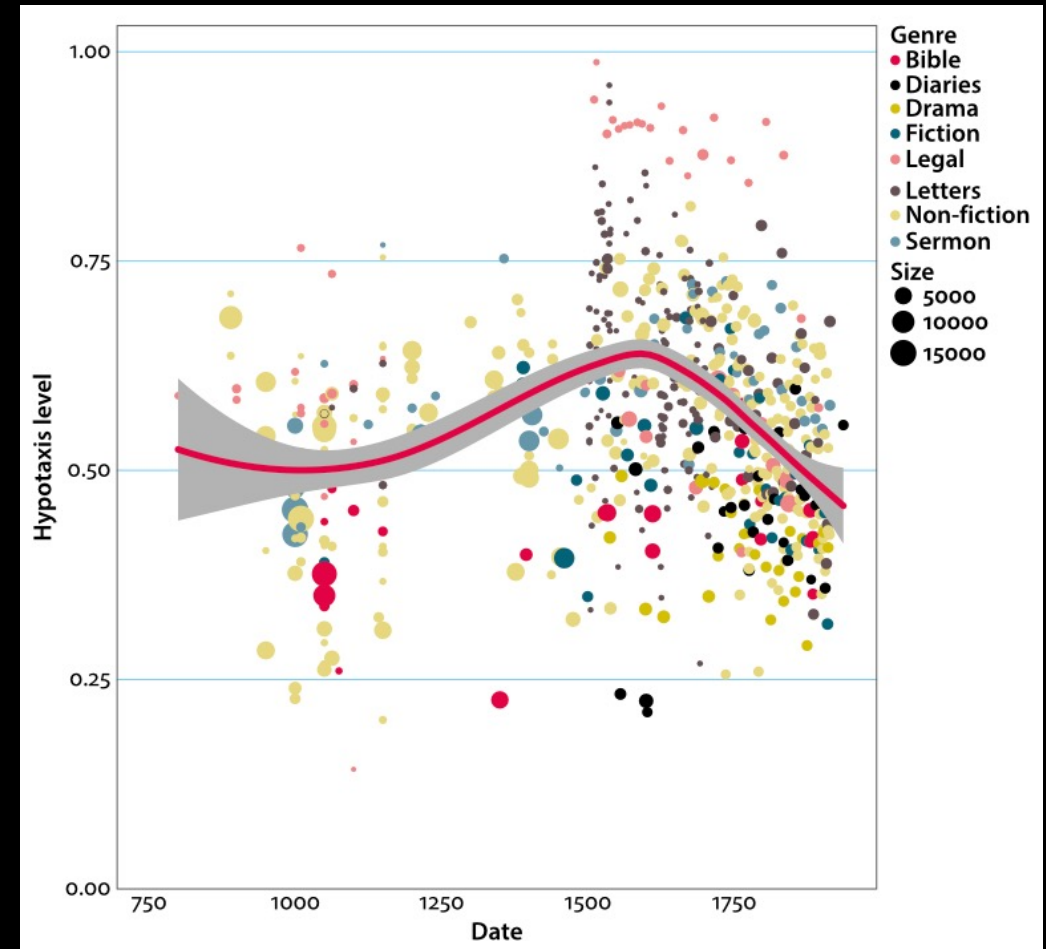
- Wallenberg (2016):
 - “It is worth noting that the type of data presented ... is entirely unobtainable without diachronic parsed corpora (i.e. treebanks), ones that have been annotated at a very high level of accuracy (i.e. hand-corrected by expert syntacticians, not just automatically parsed).” (2016: e239)
 - Analysis:
 - relative clauses are ambiguous between two syntactic analyses, **matching** and **raising** (Sauerland 2003)
 - the matching analysis (which allows for extraposition) is losing ground over time
 - Speculatively: a change so slow that it actually started in Proto-Indo-European?

CASE STUDY 3: PARATAXIS AND HYPOTAXIS

- Common narrative in historical linguistics: **parataxis** (loose association of clauses) becomes replaced by **hypotaxis** (embedding) over time.
 - Parataxis: *The road was icy. I slipped.*
 - Hypotaxis: *Because the road was icy, I slipped.*
- Walkden (2024) investigates whether this is true for the history of English, using the YCOE, PPCME2, PPCME and PPCMBE2.
 - Independent clauses are tagged as IP-MAT.
 - Embedded clauses are tagged as IP-SUB.
- We can establish the proportion of all clauses that are embedded.

PARATAXIS AND HYPOTAXIS: RESULTS

- No general trend towards hypotaxis, contrary to the common claim
- Genre plays a major role:
 - Legal texts (statutes) have a lot of embedding
 - Bibles and diaries have very little
- There is an imbalance in what data is available to us for different periods, especially before 1500



INTERIM CONCLUSION

- Parsed corpora can be used for all kinds of things!
 - Testing theoretical predictions in historically-attested languages
 - Investigating change in progress (including frequency shifts)
 - And much more, e.g. social or geographical stratification
- Next part: getting to use parsed corpora yourself!

REFERENCES: CORPORA (1)

- Beck, Jana. 2013. Penn Parsed Corpora of Historical Greek (PPCHiG). <https://www.ling.upenn.edu/~janabeck/greek-corpora.html>.
- Booth, Hannah M., Anne Breitbarth, Aaron Ecaj & Melissa Farasyn. (2020). A Penn-style Treebank of Middle Low German. *Proceedings of the 12th Conference on Language Resources and Evaluation (LREC 2020)*: 766–75.
- Diller, Hans-Jürgen, Hendrik De Smet & Jukka Tyrkkö. 2011. A European database of descriptors of English electronic texts. *The European English Messenger* 19. 21–35.
- Faggionato, Christian, and Marieke Meelen. 2019. Developing the Old Tibetan Treebank. *Proceedings of Recent Advances in Natural Language Processing, Varna, Bulgaria*, 304–312.
- Galves, Charlotte, Aroldo Leal de Andrade and Pablo Faria. 2017. Tycho Brahe Parsed Corpus of Historical Portuguese. <https://www.tycho.iel.unicamp.br/corpus/>.
- Gotthard, Lisa. 2022–. The Parsed Corpus of Scottish Correspondence (PCSC).
- Kroch, Anthony, and Beatrice Santorini. 2021. Penn-BFM Parsed Corpus of Historical French, version 1.0. <https://github.com/beatrice57/mcvf-plus-ppchf>.

REFERENCES: CORPORA (2)

- Kroch, Anthony, Beatrice Santorini, and Lauren Delfs. 2004. The Penn-Helsinki Parsed Corpus of Early Modern English (PPCEME). Department of Linguistics, University of Pennsylvania. CD-ROM, first edition, release 3, <http://www.ling.upenn.edu/ppche-release-2016/PPCEME-RELEASE-3>.
- Kroch, Anthony, Beatrice Santorini, and Ariel Diertani. 2016. The Penn Parsed Corpus of Modern British English (PPCMBE2). Department of Linguistics, University of Pennsylvania. CD-ROM, second edition, release 1, <http://www.ling.upenn.edu/ppche-release-2016/PPCMBE2-RELEASE-1>.
- Kroch, Anthony, and Ann Taylor. 2000. The Penn-Helsinki Parsed Corpus of Middle English (PPCME2). Department of Linguistics, University of Pennsylvania. CD-ROM, second edition, release 4, <http://www.ling.upenn.edu/ppche-release-2016/PPCME2-RELEASE-4>.
- Kytö, Merja. 1996. Manual to the Diachronic Part of The Helsinki Corpus of English Texts: Coding Conventions and Lists of Source Texts. 3rd edition. <http://korpus.uib.no/icame/manuals/HK/INDEX.HTM>.
- Lash, Elliott. 2014. The Parsed Old and Middle Irish Corpus (POMIC). Version 0.1. https://www.dias.ie/index.php?option=com_content&view=article&id=6586&Itemid=224&lang=en.
- Li, Man. 2017. Chinese Parsed Historical Corpus (ChiParHC).

REFERENCES: CORPORA (3)

- Martineau, France, Paul Hirschbühler, Anthony Kroch, and Yves Charles Morin. 2021. MCVF Corpus, parsed, version 2.0. <https://github.com/beatrice57/mcvf-plus-ppchf>.
- Martins, A. M. 2000—. CORDIAL-SIN: Corpus Dialectal para o Estudo da Sintaxe / Syntax-oriented Corpus of Portuguese Dialects. Lisboa, Centro de Linguística da Universidade de Lisboa. <http://www.clul.ulisboa.pt/en/10-research/314-cordial-s>.
- Meelen, Marieke, & David W. E. Willis. 2022. Towards a historical treebank of Middle and Modern Welsh: Syntactic parsing. *Journal of Historical Syntax* 6 (5), 1–32.
- Petré, Peter, Lynn Anthonissen, Sara Budts, Enrique Manjavacas, Emma-Louise Silva, William Standing and Odile A.O. Strik. 2019. Early Modern Multiloquent Authors (EMMA): Designing a large-scale corpus of individuals' languages. *ICAME Journal* 43 (1), 83–122.
- Pintzuk, Susan, and Leendert Plug. 2002. The York-Helsinki Parsed Corpus of Old English Poetry. Department of Linguistics, University of York. Oxford Text Archive, first edition, <http://www-users.york.ac.uk/~lang18/pcorpus.html>.
- Santorini, Beatrice. 2021. Penn Parsed Corpus of Historical Yiddish, v1.0. <https://github.com/beatrice57/penn-parsed-corpus-of-historical-yiddish>.
- Taylor, Ann, Mitch Marcus and Beatrice Santorini. 2003. The Penn Treebank: an overview. In Anne Abeillé (ed.), *Treebanks*, 5–22. Dordrecht: Springer.

REFERENCES: CORPORA (4)

- Taylor, Ann, Arja Nurmi, Anthony Warner, Susan Pintzuk, and Terttu Nevalainen. 2006. The York-Helsinki Parsed Corpus of Early English Correspondence (PCEEC). Department of Linguistics, University of York. Oxford Text Archive, first edition, <http://www-users.york.ac.uk/~lang22/PCEEC-manual/index.htm>.
- Taylor, Ann, Anthony Warner, Susan Pintzuk, and Frank Beths. 2003. The York-Toronto-Helsinki Parsed Corpus of Old English Prose (YCOE). Department of Linguistics, University of York. Oxford Text Archive, first edition, <http://www-users.york.ac.uk/~lang22/YcoeHome1.htm>.
- Tortora, Christina, Beatrice Santorini, Frances Blanchette, & C.E.A. Diertani. 2017. *The Audio-Aligned and Parsed Corpus of Appalachian English* (AAPCAppE), version 0.1. URL: www.aapcappe.org
- Truswell, Robert, Rhona Alcorn, James Donaldson and Joel Wallenberg. 2019. A Parsed Linguistic Atlas of Early Middle English. In Rhona Alcorn, Joanna Kopaczyk, Bettelou Los and Benjamin Molineaux (eds.), *Historical dialectology in the digital age*, 17–38. Edinburgh: Edinburgh University Press.
- Wallenberg, Joel C., Anton Karl Ingason, Einar Freyr Sigurðsson and Eiríkur Rögnvaldsson. 2011. Icelandic Parsed Historical Corpus (IcePaHC). Version 0.9. http://www.linguist.is/icelandic_treebank
- Zimmermann, Richard. 2014–a. Parsed Corpus of Middle English Poetry (PCMEP).
- Zimmermann, Richard. 2014–.b Geneva Corpus of Early German (GeCEG).

REFERENCES: OTHER LITERATURE (1)

- Biberauer, Theresa, Anders Holmberg and Ian Roberts. 2014. A syntactic universal and its consequences. *Linguistic Inquiry* 45 (2), 169–225.
- Fillmore, Charles J. 1992. “Corpus linguistics” or “Computer-aided armchair linguistics”. In Jan Svartvik (ed.), *Directions in corpus linguistics*, 35–60. Berlin: Mouton de Gruyter.
- Gries, Stefan Th. 2010. Corpus linguistics and theoretical linguistics: a love-hate relationship? Not necessarily... *International Journal of Corpus Linguistics* 15, 327–343.
- Hejná, Míša, & George Walkden. 2022. *A history of English*. Berlin: Language Science Press.
- McEnery, Tony, & Andrew Wilson. 2001. *Corpus linguistics*. 2nd edition. Edinburgh: Edinburgh University Press.
- Miller, Thomas. 1890. *The Old English version of Bede’s Ecclesiastical History of the English People*. London: Trübner.

REFERENCES: OTHER LITERATURE (2)

- Miller, Thomas. 1898. *A Collation of Four MSS. of the Old English Version of Bede's Ecclesiastical History of the English People*. London: Kegan Paul, Trench & Trübner.
- Pintzuk, Susan, & Ann Taylor. 2006. The loss of OV order in the history of English. In Ans van Kemenade & Bettelou Los (eds.), *The handbook of the history of English*, 249–278. Oxford: Blackwell.
- Sauerland, Uli. 2003. Unpronounced heads in relative clauses. In Kerstin Schwabe and Susanne Winkler (eds.), *The interfaces: Deriving and interpreting omitted structures*, 205–226. Amsterdam: John Benjamins.
- Sheehan, Michelle, Theresa Biberauer, Ian Roberts and Anders Holmberg. 2017. *The Final-over-Final Condition: a syntactic universal*. Cambridge, MA: MIT Press.
- Stefanowitsch, Anatol. 2020. *Corpus linguistics: a guide to the methodology*. Berlin: Language Science Press.

REFERENCES: OTHER LITERATURE (3)

- Taylor, Ann. 2020. Treebanks in historical syntax. *Annual Review of Linguistics* 6, 195–212.
- Walkden, George. 2014. Object position and Heavy NP Shift in Old Saxon and beyond. In Kristin Bech & Kristine G. Eide (eds.), *Information structure and syntactic change in Germanic and Romance languages*, 313–340. Amsterdam: John Benjamins.
- Walkden, George. 2024. Parataxis and hypotaxis in the history of English. In Thijs Porck, Moragh S. Gordon and Luisella Caon (eds.), *Keys to the History of English: Diachronic Linguistic Change, Morpho-Syntax and Lexicography*, 10–33. Amsterdam: John Benjamins.
- Wallenberg, Joel C. 2016. Extraposition is disappearing. *Language (Historical Syntax)* 92 (4), e237–e256.
- Wallenberg, Joel C., Henri Kauhanen, George Walkden & Caroline Heycock. **2026**. Grammar competition and variational learning. *The Wiley-Blackwell Companion to Diachronic Linguistics*. Oxford: Wiley-Blackwell.